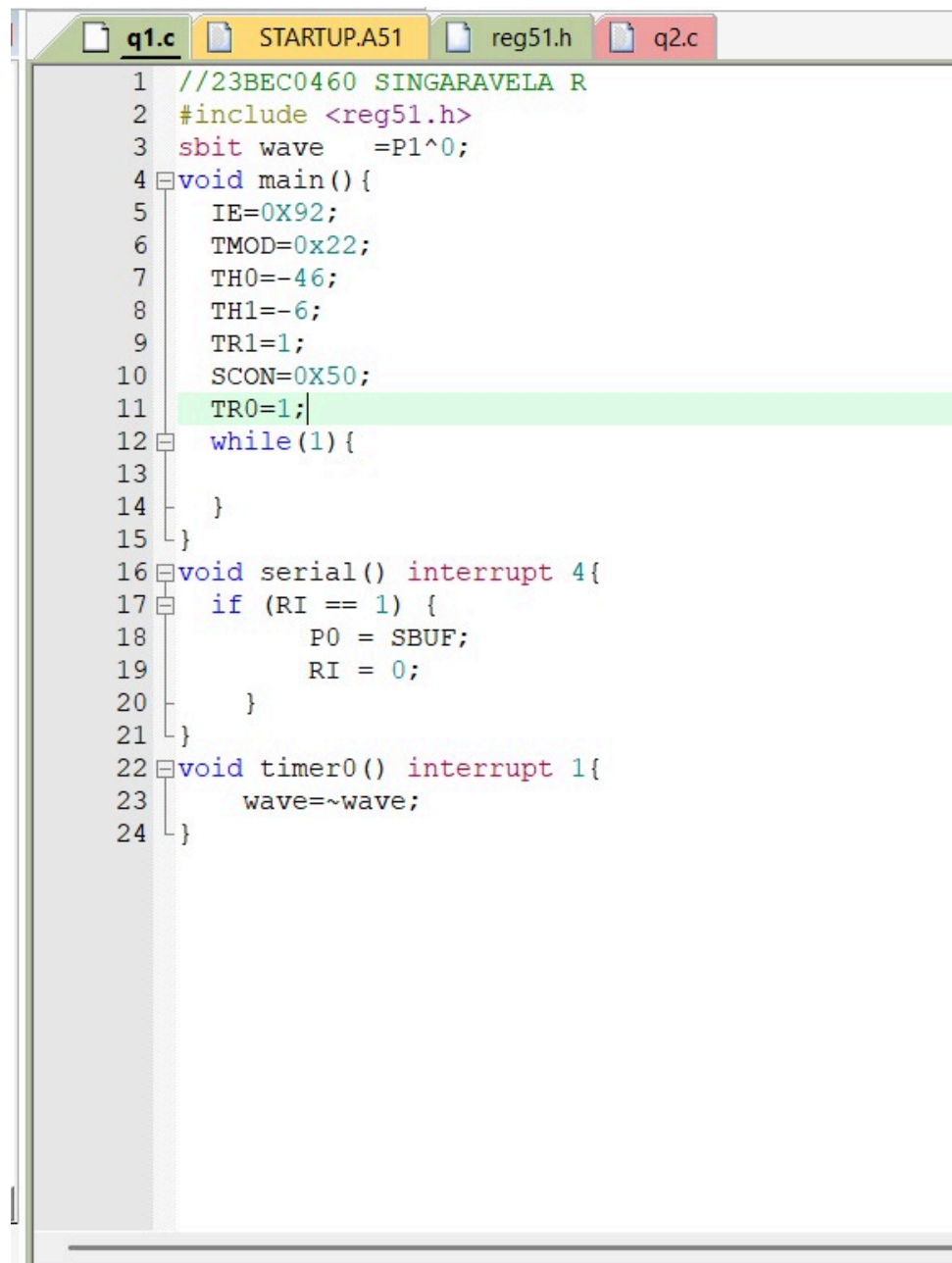


Embedded C Lab DA 5

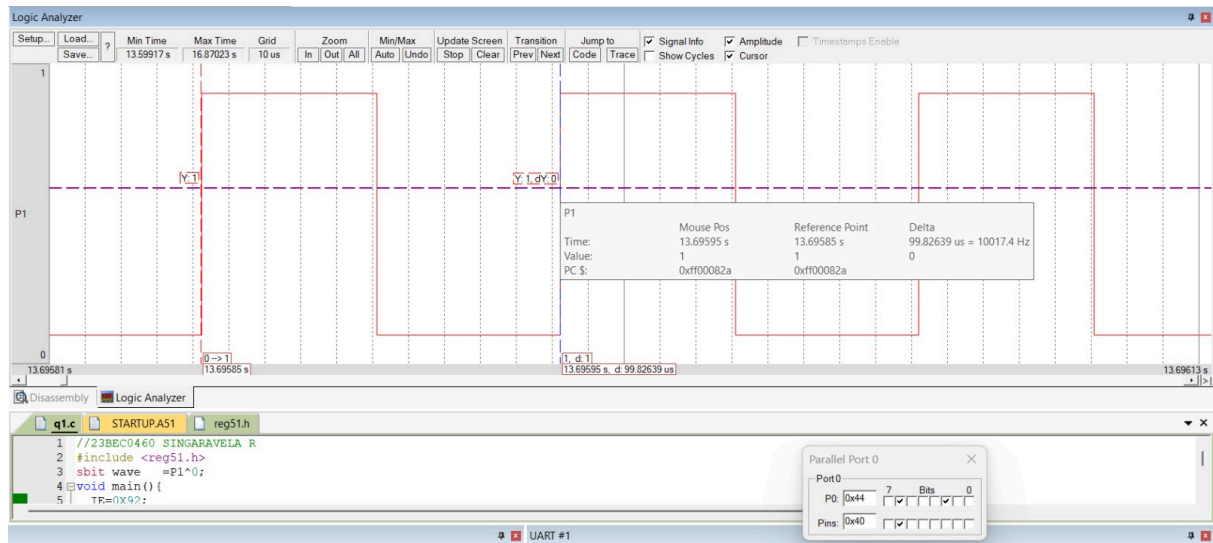
Singara vela R
23BEC0460

Write a 8051 based Embedded C program using interrupts to do the following:
a) Receive data serially and sent it to P0, b) Make timer 0 generate a square wave of 100micro second time period on P1.0. Assume that XTAL=11.0592. Set the baud rate at 4800.



```
1 //23BEC0460 SINGARAVELA R
2 #include <reg51.h>
3 sbit wave =P1^0;
4 void main(){
5     IE=0X92;
6     TMOD=0x22;
7     TH0=-46;
8     TH1=-6;
9     TR1=1;
10    SCON=0X50;
11    TR0=1;
12    while(1){
13
14    }
15 }
16 void serial() interrupt 4{
17     if (RI == 1) {
18         P0 = SBUF;
19         RI = 0;
20     }
21 }
22 void timer0() interrupt 1{
23     wave=~wave;
24 }
```

Output:



Code :

```
//23BEC0460 SINGARAVELA R
```

```
#include <reg51.h>
```

```
sbit wave = P1^0;
```

```
void main(){
```

```
    IE=0X92;
```

```
    TMOD=0x22;
```

```
    TH0=-46;
```

```
    TH1=-6;
```

```
    TR1=1;
```

```
    SCON=0X50;
```

```
    TR0=1;
```

```
    while(1){
```

```
    }
```

```
}
```

```
void serial() interrupt 4{
```

```
    if (RI == 1) {
```

```
        P0 = SBUF;
```

```
        RI = 0;
```

```
    }
```

```
}
```

```
void timer0() interrupt 1{
```

```
    wave=~wave;
```

```
}
```

Calculation:

Task 5A-

Square wave delay

calculation:

$$\frac{2 \times 100 \times 10^{-6}}{1.032 \times 10^{-6} \times 2} = 189.46$$

$$65535 - 189 = 65346$$

$$TH0 = -46$$

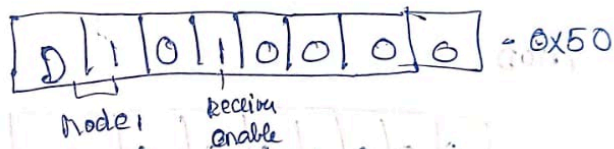
$$= FF49D1$$

$$TH0 = FF$$

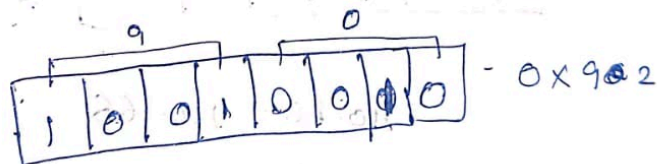
$$TLO = 48$$

Serial Communication: TH1 = -6

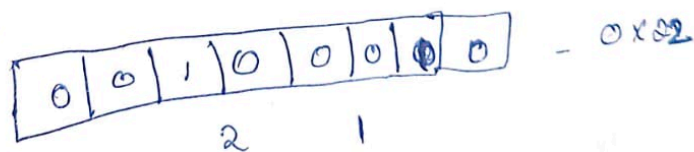
SCON:



IE:



THOD:



2. Write a 8051 based Embedded C program using timer 0 to generate a 500 Hz and timer 1 to 7kHz square wave frequency on P1.0 and P1.1 respectively using Interrupts. Then examine the frequency using the KEIL IDE inbuilt Logic Analyzer.

Code:

```
//23BEC0460 SINGARAVELA R
#include <reg51.h>
sbit P1_0 = P1^0;
sbit P1_1 = P1^1;

void timer0_ISR(void) interrupt 1 {
    TH0 = 0xFC;
    TL0 = 0x66;
    P1_0 = ~P1_0;
}

void timer1_ISR(void) interrupt 3 {
    TH1 = 0xFF;
    TL1 = 0xBE;
    P1_1 = ~P1_1;
}

void main(void) {
    TMOD = 0x11;

    TH0 = 0xFC;
    TL0 = 0x66;

    TH1 = 0xFF;
    TL1 = 0xBE;

    ET0 = 1;
    ET1 = 1;
    EA = 1;

    TR0 = 1;
    TR1 = 1;

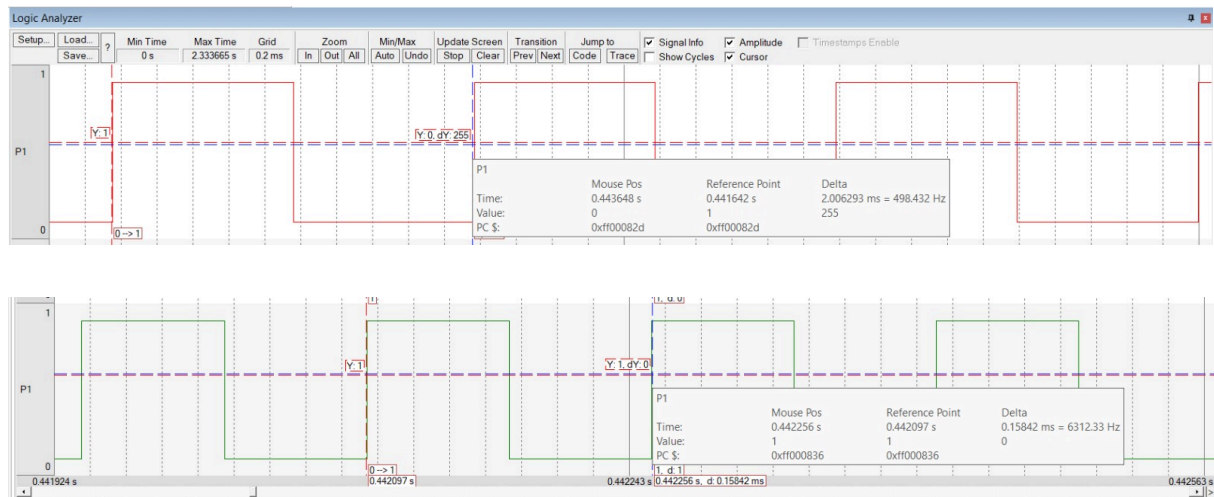
    while(1);
}
```

```

1 //23BEC0460 SINGARAVELA R
2 #include <reg51.h>
3 sbit P1_0 = P1^0;
4 sbit P1_1 = P1^1;
5
6 void timer0_ISR(void) interrupt 1 {
7     TH0 = 0xFC;
8     TL0 = 0x66;
9     P1_0 = ~P1_0;
10 }
11
12 void timer1_ISR(void) interrupt 3 {
13     TH1 = 0xFF;
14     TL1 = 0xBE;
15     P1_1 = ~P1_1;
16 }
17
18 void main(void) {
19     TMOD = 0x11;
20
21     TH0 = 0xFC;
22     TL0 = 0x66;
23
24     TH1 = 0xFF;
25     TL1 = 0xBE;
26
27     ET0 = 1;
28     ET1 = 1;
29     EA = 1;
30
31     TR0 = 1;
32     TR1 = 1;
33
34     while(1);
35 }

```

Output:



Calculation:

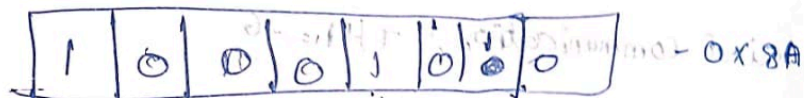
Task 5B:

Square wave
calculation:

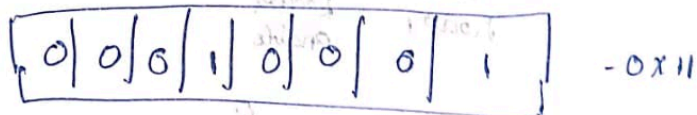
$$500 \text{ Hz} = \frac{1}{2 \times 500 \times 1.085 \times 10^{-6}} = 921 \Rightarrow 65535 - 921 = 64614 = \text{FC66}$$

$$2 \text{ kHz} = \frac{1}{2 \times 7 \times 10^3 \times 1.085 \times 10^{-6}} = 66 \Rightarrow 65535 - 66 = 65469 = \text{FFBD}$$

IE:



TMOD:



TH0 = FC TL0 = 66

TH1 = FF TL1 = BD